AP PHYSICS 1

Unit 1 • Self Check 1.1 • Topics 1.1 to 1.3 • Answer key

Self Check 1.1 • Topics 1.1 to 1.3

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Question 1

Which pair contains only vector quantities?

(A) Displacement and velocity
(B) Speed and velocity
(C) Distance and acceleration
(D) Time and force

Solution

Answer: (A). Displacement and velocity both have a direction. Speed, distance and time are scalars.

Question 2

A cyclist rides 6 km east, then 2 km west, in 0.5 h. What is the average velocity?

(A) 16 km/h east
(B) 8 km/h east
(C) 8 km/h west
(D) 4 km/h east

Solution

Answer: (B). Displacement 4 km east, divided by 0.5 h: 8 km/h east. (The average speed is 16 km/h.)

Question 3

A car moving in the negative x-direction is slowing down. Which signs are correct?

(A) $v < 0, a < 0$
(B) $v > 0, a < 0$
(C) $v < 0, a > 0$
(D) $v > 0, a > 0$

Solution

Answer: (C). Moving in the negative direction: $v < 0$. Slowing down: a has the opposite sign, $a > 0$.

Question 4

An object starts from rest and accelerates at $4\,\text{m/s}^2$ over a distance of $18\,\text{m}$. What is its final speed?

(A) $6\,\text{m/s}$
(B) $72\,\text{m/s}$
(C) $144\,\text{m/s}$
(D) $12\,\text{m/s}$

Solution
Answer: (D). $v^2 = 0 + 2(4)(18) = 144$, so $v = 12\,\text{m/s}$.

Question 5

On a velocity-time graph, the area between the graph and the time axis represents the

(A) acceleration
(B) displacement
(C) speed
(D) time

Solution
Answer: (B). Velocity $\times$ time $=$ displacement.

Question 6

The position-time graph of an object is a horizontal line. The object is

(A) at rest
(B) moving with constant acceleration
(C) moving at a constant nonzero velocity
(D) slowing down

Solution
Answer: (A). The slope (velocity) is zero.

Question 7 (Free response)

An elevator starts from rest and moves up. It reaches $3\,\text{m/s}$ in $2\,\text{s}$, moves at $3\,\text{m/s}$ for $4\,\text{s}$, then stops in $3\,\text{s}$.

(a) Find the acceleration during each part.
(b) Find the total displacement.
(c) Find the average velocity for the whole trip.

Solution
(a) $1.5\,\text{m/s}^2$; 0; $-1\,\text{m/s}^2$. (b) $\frac{1}{2}(2)(3) + 4(3) + \frac{1}{2}(3)(3) = 3 + 12 + 4.5 = 19.5\,\text{m}$ up. (c) $19.5/9 \approx 2.2\,\text{m/s}$ up.